AI INFRASTRUCTURE / EXPERIMENTAL TECHNICAL REPORT

Memory-Bounded Teacher Scoring

Contract-preserving FSDP integration, allocator-bounded capacity and direct Worker-to-Student evidence

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Document status	AI-assisted technical draft; all new content pending Human review
Evidence snapshot	Base `0fd56a6`, with the subsequent recorded Worker and mask checks
Verified scope	Single-GPU Qwen2/FSDP2; bounded numerical, capacity and restricted-update experiments

Review boundary. This is a new report. It does not inherit the author-review status of the earlier `c1ab1f4` report editions. Those historical reports and the 30 October Markdown/PDF supplement remain separate snapshots. Section 10 reports the subsequent bounded Worker and mask checks with their actual measurement limits.

Measured allocation is reported with its allocator and execution scope. Row chunking is established prior art; this report claims no new selected-logit kernel or general latency improvement.

Abstract

A compact teacher response can still require an entire token-by-vocabulary logits allocation. This implementation moves row chunking before a Qwen2 LM head while retaining full-vocabulary-normalized scores at Student candidate IDs and the Teacher's own Top-K IDs. The adapter runs inside the ordinary outer FSDP forward, preserves module hooks and tied parameters, and uses native dense forward when an unbudgeted tile covers every projected row. A controlled single-GPU factorial study completed 384 direct-model calls and 64 actual Worker calls without reproducing an earlier CUDA abort. Under a PyTorch allocator fraction of 0.25, the offload-on dense path passed at 1536 rows and failed at 2048, while chunked execution reached 32768; same-length dense correctness was checked only through 8192. At fraction 0.50, dense passed at 8704 and failed at 9216; a higher-budget dense reference confirmed the chunked 9216-row output. Actual Worker outputs also preserved the pinned loss, checked gradients and one final-RMSNorm AdamW update. A later real Worker check matches the complete compact output at 8192 rows, and at 9216 against a higher-budget reference. A separate no-fraction 1024-row check measures memory, not maximum capacity. Full RL training and Ray dispatch remain outside the recorded evidence.

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1 Motivation and contribution boundary

Distillation consumers require different information: selected logits, conditional probabilities, full-vocabulary log-probabilities and Teacher Top-K are distinct contracts. An interface returning only NK values does not imply that its implementation avoids an NV projection or allocation. Here N is the number of model rows actually projected, V vocabulary size and K candidate width; padded rows count toward N.

Existing per-row scoring interfaces and indexed selected-logit primitives already address parts of this problem. The earlier source survey found relevant SGLang and vLLM interfaces and IndexedLogits; those observations remain tied to their recorded revisions.[1, 3, 5, 7] The engineering subject of this report is narrower: preserving the complete pinned verl Teacher contract while bounding the logits tile in its Qwen2/FSDP execution. Row chunking is an established method, and the selected-logit-only fusion proposal remains **NO-GO** for this complete contract.

The target is verl PR #6920 at the fixed revision `87cd055e872f9fdd7b976fcd797a0934b6dba265`. The patch is opt-in; its disabled branch retains the upstream behavior. Source observations and the present local runs do not establish upstream acceptance.

2 Complete scoring and consumer contract

For row t, let hidden state h_t, head weight W, bias b, and candidate support C_t define:

$$z_{t,i} = h_t^{\mathsf{T}} W_i + b_i, \tag{1}$$

$$\ell^{\text{full}}_{t,i} = z_{t,i} - \mathrm{LSE}_{j \in \mathcal{V}}\, z_{t,j}, \tag{2}$$

$$\ell^{C}_{t,i} = z_{t,i} - \mathrm{LSE}_{j \in C_t}\, z_{t,j} = \ell^{\text{full}}_{t,i} - \log Q_t(C_t). \tag{3}$$

The returned candidate values are $\ell^{\text{full}}_{t,i}$, not $\ell^{C}_{t,i}$. The second output is the Teacher's own Top-K over its complete vocabulary. Omitting unselected logits prevents reconstruction of the full normalizer in general; it also fails to determine Teacher Top-K. Computing only the Student candidate projection therefore cannot substitute for this two-output interface.

C_t denotes a mathematical support set in these equations. The actual wire carries ordered int64 IDs with shape $[B, T, K]$, can contain repeated IDs, and must preserve that order and multiplicity in gathered scores. Candidate IDs already address causal-logit rows. The adapter adds neither a next-token shift nor a temperature transformation.

The patched Worker still returns exactly `teacher_on_student_logp` and `teacher_topk_ids` as detached CPU nested tensors in the original valid-row order and with unchanged offsets. Padded candidate IDs are scattered using the actual prepared attention mask and valid rows are restored through that same mask. Multi-sequence padded inputs are supported by this layout path; multi-sequence packed/remove-padding attention is rejected pending verified varlen execution.

For context, support-normalized reverse KL has the fixed-support form:

$$p^C_i = \frac{e^{z^S_i/\tau_S}}{\sum_{j\in C} e^{z^S_j/\tau_S}}, \qquad q^C_i = \frac{e^{z^T_i/\tau_T}}{\sum_{j\in C} e^{z^T_j/\tau_T}}, \tag{4}$$

$$L_C = \sum_{i\in C} p^C_i \log \frac{p^C_i}{q^C_i}, \tag{5}$$

$$\frac{\partial L_C}{\partial z^S_j} = \frac{p^C_j}{\tau_S}\left(\log \frac{p^C_j}{q^C_j} - L_C\right), \quad j \in C. \tag{6}$$

This analytical expression explains why the full normalizer cancels after conditional renormalization. It is not a declaration that all OPD losses, clamps or masks implement the same objective. Conditional KL omits

support mass and tail structure:

$$D_{\mathrm{KL}}(P\|Q) = D_{\mathrm{Ber}}(\alpha\|\beta) + \alpha D_{\mathrm{KL}}(P^C\|Q^C) + (1-\alpha)D_{\mathrm{KL}}(P^{\bar{C}}\|Q^{\bar{C}}). \tag{7}$$

The actual consumer check in Section 8 uses the pinned `compute_reverse_kl_topk` source, including its configured normalization and clamps, rather than replacing it with this simplified formula. The original source survey distinguished this objective from other OPD reward and subset-loss constructions.[2, 4, 6]

3 FSDP execution and bounded readout

3.1 Module and hook lifecycle

The adapter is installed on the ordinary unwrapped `Qwen2ForCausalLM` before FSDP wrapping. It retains the parameter objects, module tree and tied input/output weight. A scored request enters `self.engine.module(...)` through the ordinary outer FSDP call. Inside that call, the backbone and final norm run once; flattened hidden rows then enter `self.lm_head(tile)` as a module for each tile. Calling the module preserves its own pre/post hooks; directly multiplying its weight would bypass that lifecycle.

For a tile of at most B rows, logits are promoted to FP32 for vocabulary log-sum-exp and candidate gather; scores are cast back to the logits dtype and Teacher Top-K IDs are retained. Tiles are discarded without concatenating full logits. Compact output storage remains proportional to NK; the backbone hidden state and model/FSDP allocations remain part of the complete forward.

The verified target is tied-embedding Qwen2 with an ordinary dense linear head, BF16, world size one, eval/no-grad and no KV cache. Adapter guards reject unsupported multi-rank, untied-head, LoRA, quantized/QAT, Liger/fused alternative-forward and sequence-parallel configurations. FP32 code paths are not equivalent to a verified FP32 GPU experiment. FSDP1, TP and multi-GPU support remain `NOT_RUN`.

3.2 CPU offload and dense fallback

FSDP2 CPU offload changes parameter residency and can dominate complete-forward timing; it does not remove the need to preserve ordinary FSDP forward boundaries. The controlled study inspects actual root/decoder offload policies instead of inferring them from a command-line flag.

With no explicit head budget, a selected tile covering all projected rows invokes `dense_full_row_fallback`: native dense forward returns first, and the existing gather/reduction runs outside FSDP. A full-row tile reduced while parameters remain resident can increase the peak; at $N = 512$, a full-row chunk of 512 increased an earlier absolute peak by about 33.7%. This observation motivated the native fallback. Any explicit head budget disables that automatic fallback, so a requested estimate is not silently replaced by unbounded dense execution. Errors and OOM propagate; the outer forward is not retried.

3.3 Budget estimate and its limits

Let s be logits element bytes, $\gamma = 1.25$ the default safety factor and H the requested head allowance in bytes. The implemented estimate is

$$M_{\mathrm{compact}} = NK(s+8), \tag{8}$$

$$m_{\mathrm{row}} = \lceil \gamma\,[V(s+8) + 24K] \rceil\,, \tag{9}$$

$$\widehat{M}(B) = M_{\mathrm{compact}} + Bm_{\mathrm{row}}, \tag{10}$$

$$B = \min\left(N, B_{\mathrm{requested}}, \left\lfloor \frac{H - M_{\mathrm{compact}}}{m_{\mathrm{row}}} \right\rfloor\right). \tag{11}$$

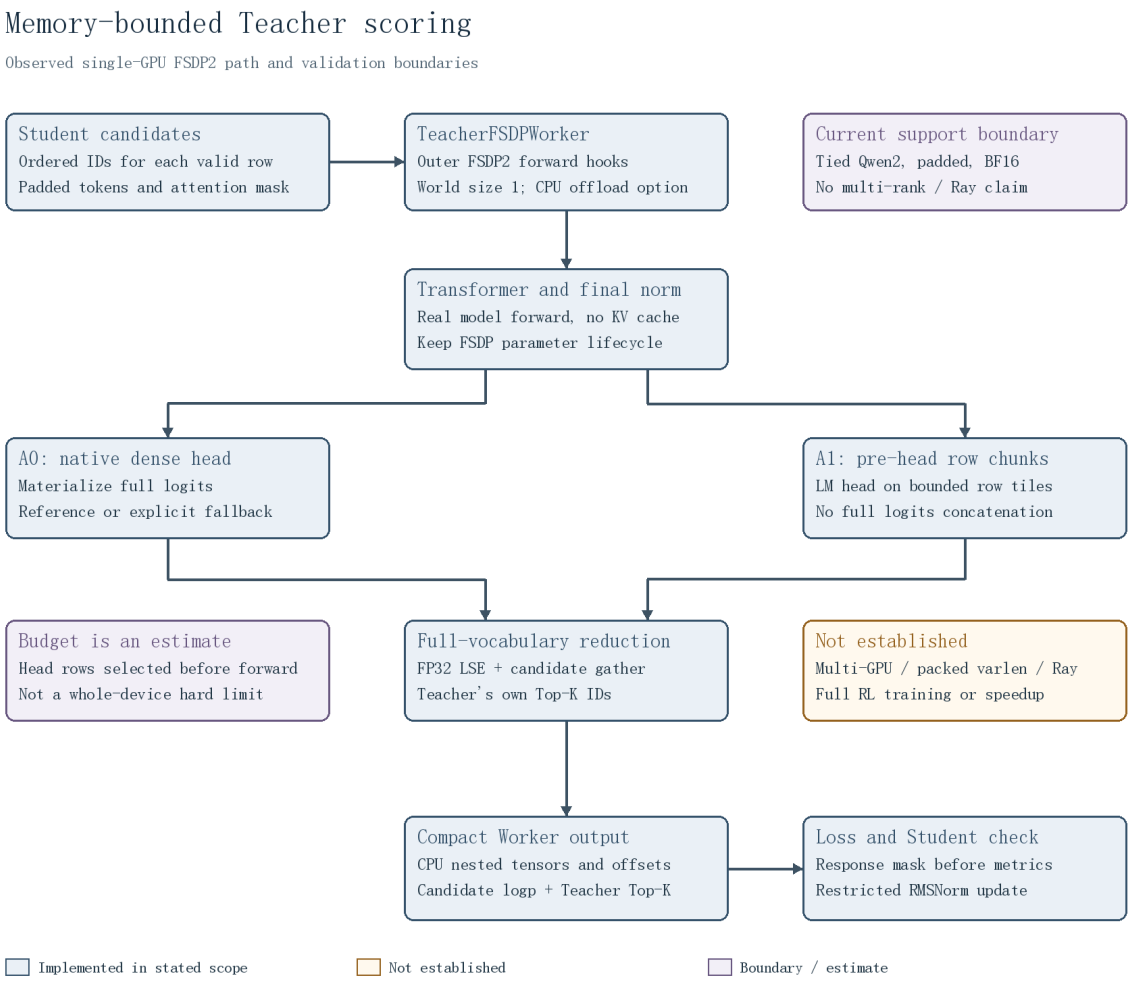


Figure 1: Memory-bounded Teacher execution and the separate direct Worker-to-Student validation path. Numerical and runtime scope is given by the experiment sections; the figure is not evidence of unrun dispatch or training paths.

The estimate includes compact output storage, a logits tile, up to two FP32 vocabulary-sized work tensors and conservative Top-K bookkeeping. If one estimated row plus compact outputs cannot fit, admission fails before the forward. The cap is on this readout estimate, not full-model memory, FSDP residency, fragmentation, other processes or all CUDA workspaces. It leaves full-vocabulary projection work approximately $2NVd$; smaller tiles trade allocation against additional head calls.

An earlier separate $N = 1024, K = 32$ FSDP2 probe selected 35/70/141 rows for 64/128/256 MiB estimated budgets. Observed readout allocated increments were approximately 51.03/101.75/204.64 MiB; complete-forward peaks were about 552.28 MiB. These observations calibrate that workload only. A head estimate and `torch.cuda.set_per_process_memory_fraction` constrain different quantities.

4 Experimental design and execution identity

Item	Fixed recorded condition
GPU	NVIDIA RTX 5070 Laptop; 8151 MiB
Model	Qwen2.5-0.5B-Instruct; BF16; tied embeddings
Model revision	`7ae557604adf67be50417f59c2c2f167def9a775`
Framework	PyTorch 2.14.1+cu130; Transformers 5.19.0
Distributed path	FSDP2, world size one; offload condition recorded separately
Input	Synthetic tokenized sequences and fixed candidate IDs; no sampled RL trajectories
Evidence source	Bilingual 30 October stage supplement and earlier bounded-adapter records

Table 1: Execution identity for the recorded experiments.

Capacity points use separate new processes, SDPA, $K = 32, B = 256$, and a single sequence. The factorial fixture instead uses padded lengths 64/37, $K = 8, B = 16$ and eager attention. These workloads are not interchangeable. Output signatures cover shape, dtype, offsets and bytes; capacity comparisons inspect all saved compact candidates and Teacher Top-K sets. The comparator preserves unresolved tie/mismatch status when compact outputs alone cannot establish a full-logits boundary tie.

4.1 Worker-aligned buffer construction

The first direct-model construction differed numerically from the actual Worker. The pinned Worker explicitly calls `model.to(dtype)`, casting Qwen2 rotary frequency buffers from FP32 to BF16; the original direct path retained FP32 buffers. Matching that single construction step changed 31/32 frequency values and restored exact equality on all 808 candidate values and 101 Teacher Top-K rows in the tested fixture. This explains that path difference, not the CUDA abort. Earlier direct-model measurements remain scoped to their recorded buffer policy.

5 S1: controlled stability

The direct-model factorial crosses Dense/Chunked with CPU offload on/off. Each condition uses three fresh processes, two warmups and 30 counted calls per process. Actual root/decoder FSDP policies match the requested setting. Each completed call is checked against the same compact reference. The direct processes completed 384 checked calls and the two actual Worker processes completed another 64 with per-call hashes and offsets. The initial CUDA `unknownerror` did not recur in this bounded set. Its cause, including the contemporaneous WSL dxg observations, remains unknown. Repetition establishes only the observed finite run set, not long-term production stability.

Entry	Condition	Processes	Calls
Direct Qwen/FSDP2	Dense, offload on	3	96
Direct Qwen/FSDP2	Chunked, offload on	3	96
Direct Qwen/FSDP2	Dense, offload off	3	96
Direct Qwen/FSDP2	Chunked, offload off	3	96
Actual verl Worker	Dense, offload on	1	32
Actual verl Worker	Chunked, offload on	1	32

Table 2: Completed calls include warmups: $3(2 + 30) = 96$ per direct condition.

6 Capacity under stated allocator fractions

The 0.25 fraction corresponds to a nominal 2,129,510,400-byte PyTorch caching-allocator allowance on this device. It is not an all-device hard partition. The boundary scan narrows tested brackets at 512-token resolution; success/OOM points do not prove monotonicity or an exact maximum. At fraction 0.25,

Fraction	Offload	Dense pass / next OOM N	Chunked largest tested runnable N	Dense-checked chunked through N
0.25	On	1536 / 2048	32768	8192
0.25	Off	512 / 1024	16384 / OOM 16896	4096
0.50	On	8704 / 9216	32768	9216

Table 3: Observed grid points and separate numerical-equivalence boundaries.

offload-on compact comparisons passed at $N = 1024, 2048, 4096, 8192$ and offload-off comparisons at $N = 2048, 4096$, with maximum logprob difference zero. Longer Dense references used fraction 0.55; reference attempts at $N = 12288$ on and $N = 8192$ off OOMed. The corrected comparator reran all six cases after adding shape/signature validation; prior comparison records were retained.

Thus the offload-on 8192/1536 observed length ratio is 5.33, while 32768 remains execution-capacity evidence without full same-N Dense correctness. At fraction 0.50, Chunked ran at 8192/16384/24576/32768. The $N = 8192$ same-budget comparison matched; a fraction-0.55 Dense reference at $N = 9216$ matched all 294,912 candidate values and all 9216 Top-K sets from fraction-0.50 Chunked. This verified grid point is 512 tokens (5.88%) beyond Dense's largest tested 8704 pass. The higher-budget reference supplies correctness, not a same-budget capacity advantage.

7 M1 and T1: memory scope and descriptive timing

7.1 Allocator peaks and whole-device observations

The lower call-only allocator peak does not erase loading/wrapping allocations. Reserved memory also depends on allocator history. The $N = 1024$ sampled maxima differ by 171 MiB; that is not an exact process-VRAM saving. At the fraction-0.50 Dense OOM, CUDA still reported approximately 3401 MiB free, identifying an allocator-budget boundary rather than device exhaustion. No unrestricted-allocator long-Worker capacity result is recorded here.

7.2 Worker-aligned paired forward timing

T1 uses $N = 1024, K = 32, B = 256$, six A/A blocks and eight interleaved ABBA/BAAB blocks per offload condition, giving 16 samples per A/B label. Loading and correctness validation are outside the

Metric, MiB	$N = 1024$, M1		$N = 8192$, fraction 0.50	
	Dense	Chunked	Dense	Chunked
Forward allocated peak	1517.09	665.31	3596.83	681.66
Process allocator peak incl. load/wrap	1517.09	958.48	3596.83	958.48
Forward reserved peak	1748	1572	3846	1572
Sampled whole-device used max	3429	3258	5504	3229
Device sample count (not MiB)	40	30	36	33

Table 4: Separate fresh Dense/Chunked processes, offload on. The 100 ms device sampler includes all processes and may miss transient peaks.

timed complete direct-model/FSDP forward; compact CPU copies are inside it. Allocated decreases were

Offload	Metric	A0 Dense	A1 Chunked
On	Allocated peak MiB	1517.09	664.97
Off	Allocated peak MiB	2467.26	1615.48
On	perf-counter p50 / p95 ms	192.580 / 193.953	192.978 / 198.150
On	raw-clock p50 / p95 ms	190.491 / 191.858	190.926 / 196.081
Off	perf-counter p50 / p95 ms	87.241 / 114.847	78.709 / 116.953
Off	raw-clock p50 / p95 ms	87.567 / 115.198	78.986 / 117.407

Table 5: Recorded direct-model paired measurements with Worker-aligned buffer casting.

56.17% on and 34.52% off in this scope. Both candidate comparisons had maximum difference zero and all 1024 Teacher Top-K sets matched. In the offload-off process, CUDA Event exceeded its surrounding perf-counter span in 30/32 A/B and 12/12 A/A samples. Earlier runs also showed host-clock and phase drift; their cause remains unresolved. Timings remain descriptive, with no general speedup claim and no production comparison across the two separate offload processes.

8 I1: actual Worker output into a restricted Student update

A real Qwen Student generates candidates once. Actual pinned `TeacherFSDPWorker` objects then execute Dense fallback and Chunked scoring sequentially. Their returned in-memory CPU nested TensorDict values are scattered into the original padded Student rows and passed directly to `compute_reverse_kl_topk`. No recorded JSON Teacher values substitute for these outputs.

The two sequences have offsets $[0, 23, 45]$, 46 padded model rows, $K = 8$ and 25 active response-prediction rows. Student selection uses temperature 0.8 followed by log-softmax and Top-K; Teacher temperature is 1. Candidates remain on their original logit rows; only the response mask receives next-token alignment. Both paths restore the same final RMSNorm weight and fresh AdamW state, learning rate 0.01 and zero weight decay. Teacher tensors are detached; support normalization, logprob clamp -10, nonnegative token-loss clamp and maximum loss clamp 10 are retained. All 360 candidate values, Teacher Top-K, mass, overlap, token losses, 6,989,056 Student logits-gradient elements, 896 RMSNorm-gradient elements and resulting updates matched exactly in the fixture. Normal outer/head hook completion, absent Teacher gradients and collection of both sequential Worker Python objects were checked. This establishes local Worker-to-loss-to-restricted-update integration; Ray dispatch, a complete Trainer, all-parameter optimization and an RL step remain `NOT_RUN`.

Observed quantity	Dense Worker	Chunked Worker
Masked scalar loss	0.0216006283	0.0216006283
Final RMSNorm gradient L2	0.0124774044	0.0124774044
Nonzero gradient elements	896	896
Changed BF16 parameter values	1	1
Parameter-update L2	0.0099792480	0.0099792480

Table 6: One final-RMSNorm AdamW update using actual Worker outputs.

9 Negative results and evidence interpretation

The initial selected-logit proposal does not satisfy full normalization plus Teacher Top-K. An earlier full-row chunk increased the complete peak and led to the explicit native fallback policy. The initial CUDA abort remains unexplained despite the bounded non-reproduction set. Buffer casting explains one direct/Worker numerical difference but provides no evidence about that abort.

An earlier project-consumer version independently cast BF16 Student logprob to FP32 on two gradient branches. Scalar loss matched upstream, but RMSNorm and logits gradients differed by about 1.07% and 1.80% relative L2. Reusing the FP32 conversion as the upstream graph does removed those differences for the fixture. This retained negative result motivates comparing gradients and updates, rather than accepting scalar-loss equality alone.

Earlier transient samples lost after WSL temporary storage was cleared are not reconstructed; later recorded runs form the cited measurement scope. The archived matrix also mislabeled a full-row head call as avoiding full logits; actual call shapes contradicted the label and the current script corrected it. Longer chunked runs beyond the Dense-reference ceiling retain `UNKNOWN` numerical equivalence. No report wording converts an exit code or a finite-output run into complete numerical or training acceptance.

10 Real long Worker, natural allocation and response-mask checks

These later measurements use the actual pinned `TeacherFSDPWorker` in fresh processes, tied Qwen2 BF16, FSDP2 CPU offload, SDPA, $K = 32$ and row chunk $B = 256$. Text and candidate IDs are synthetic. Every call records actual input, attention mask, position IDs, offsets, outer/backbone/head hooks and the complete CPU nested output. Two short processes with lengths 64/37 exercise real padding; all 3232 candidate logprobs and 101 ordered Teacher Top-K rows match. The long tests each use one causal sequence; they do not establish long multi-sequence padding behavior.

Worker condition	Allocated peak, MiB		Reserved peak, MiB	
	Dense	Chunked	Dense	Chunked
8192, both fraction 0.50	3596.638	681.630	3844	866
9216, dense 0.55 / chunked 0.50	3893.700	683.962	4144	884
1024, no fraction, A/A and B/B	1517.071	665.306	2644	1124

Table 7: Complete Worker-call allocator peaks in independent processes. The 9216 Dense fraction is a correctness reference, not a same-budget capacity comparison.

At $N = 8192$, all 262,144 candidate logprobs and all 8192 ordered Teacher Top-K rows match exactly. At $N = 9216$, all 294,912 candidate values and 9216 ordered Top-K rows match exactly across the stated fractions. Each long cell is one cold call; the frozen BF16 tolerance is also satisfied. The matching long outputs have offsets $[0, N]$. These observations do not validate 32768 against dense or establish a full Trainer run.

Without calling `set_per_process_memory_fraction`, four new $N = 1024$ processes ran in ABBA order, each with one warmup and three measured calls. The recorded actual fraction was 1.0 in each. A/A, B/B and A/B compact outputs matched, including ordered Top-K. Dense measured perf-counter samples ranged 175.754–200.623 ms and Chunked 161.573–191.955 ms; host clocks disagreed in some samples. These are descriptive small-sample timings, not an acceleration claim. The 100 ms all-device maximum samples were 4990/5120 MiB for the two Dense processes and 3584/3332 MiB for the two Chunked processes; these include other processes and can miss peaks. No no-fraction capacity ceiling was tested.

The fixed trainer's metric wrapper masks to response rows before mass, loss range and overlap aggregates; overlap advantage additionally requires positive overlap. The local diagnostic now applies the same rules while retaining raw per-token outputs. CPU tests compare with the actual fixed trainer functions and a hand-calculated oracle, including extreme or NaN prompt/padding values and zero-overlap rows; loss and gradient behavior is unchanged. A CPU replay of the prior I1 record retains the padded `teacher_mass=8.0` but excludes it: response-only Teacher mass mean is 0.732026 and maximum 0.999786 for 25 response rows. This replay is not a new GPU training step.

Two initial starts were `NOT_RUN` because the predeclared free-memory gate was not met; an initial CPU dependency import also failed and the runner was corrected before GPU execution. All ten completed Worker processes report no device-sampler failure. The subsequent probe revision makes a missing device observation fail the experiment while retaining any successful Worker output; the executed probe source and its SHA-256 remain with the raw measurements. This later error-handling correction was CPU-regression-tested and does not retroactively change the recorded runs.

11 Reproduction and final technical judgment

The base source snapshot is `0fd56a618800b5298d8c45529dda675f910c6e47`; the later Worker probe and mask regression are in the revision containing this report. The raw records bind the probe bytes used for the measured runs. The pinned loss is the actual verl FSDP distillation loss, Git blob `97d3ebf485a5ce9e1b1a435cce427d6d89284dc6`. Model and tokenizer hashes, command arguments and source identities accompany the original measurements.

The principal entries include `scripts/long_worker_probe.py`, `scripts/factorial_teacher_probe.py`, `scripts/compare_capacity_outputs.py`, `scripts/full_teacher_benchmark.py` and `scripts/worker_to_student_step.py`. Evidence identifiers include `S1_SUMMARY.json`, `capacity_runner.jsonl`, `compare_validated_summary.json`, `compare_on_n9216.json`, `T1_worker_aligned_on/off.json` and `worker_to_student_01.json`. The bilingual Markdown stage supplement supplies the preceding measured tables; the later section above uses the recorded Worker runs and CPU mask regression.

The supported conclusion is contract-preserving bounded readout with measured allocator-capacity benefits and a real restricted consumer update in the stated single-GPU environment. FP32 GPU, FSDP1, TP/multi-GPU, packed varlen, LoRA/quantization/QAT, Ray transport, representative RL trajectories, full-model training and production stability remain unverified. General latency improvement, exact total-device VRAM savings and training-quality gains are not established. All new text remains AI-assisted and pending Human review.

References

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